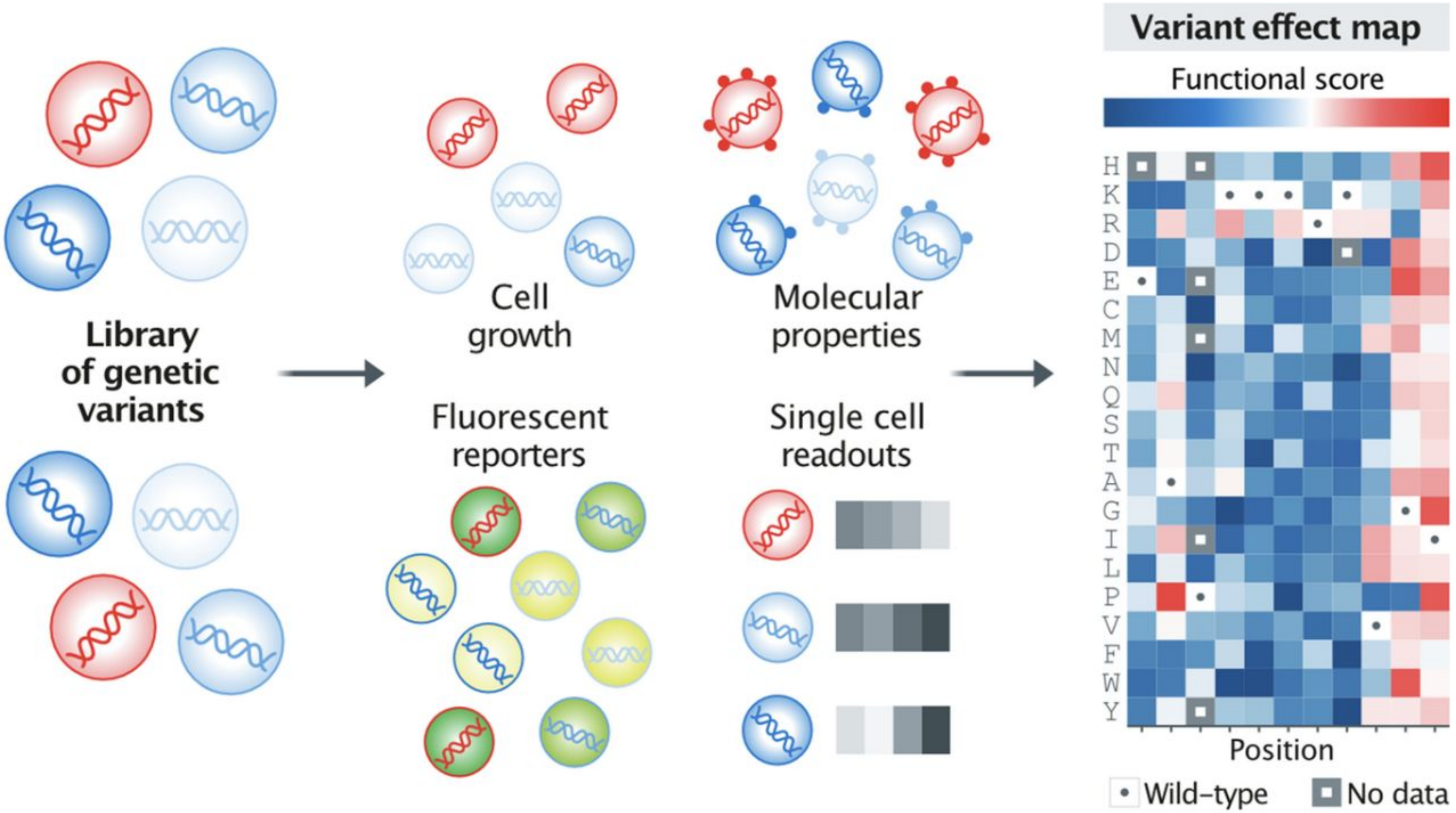
connecting people with science

Dr Alan Rubin

Why MAVEs?

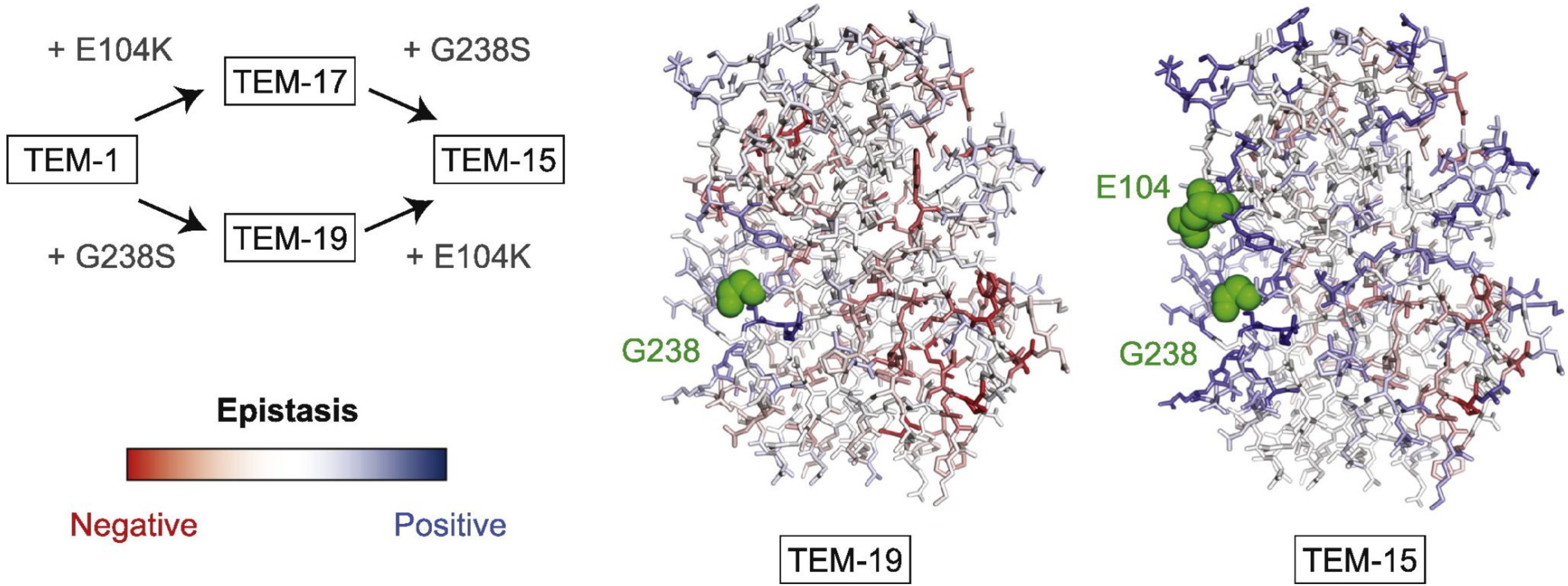
November 2025

MAVE technologies have very diverse applications



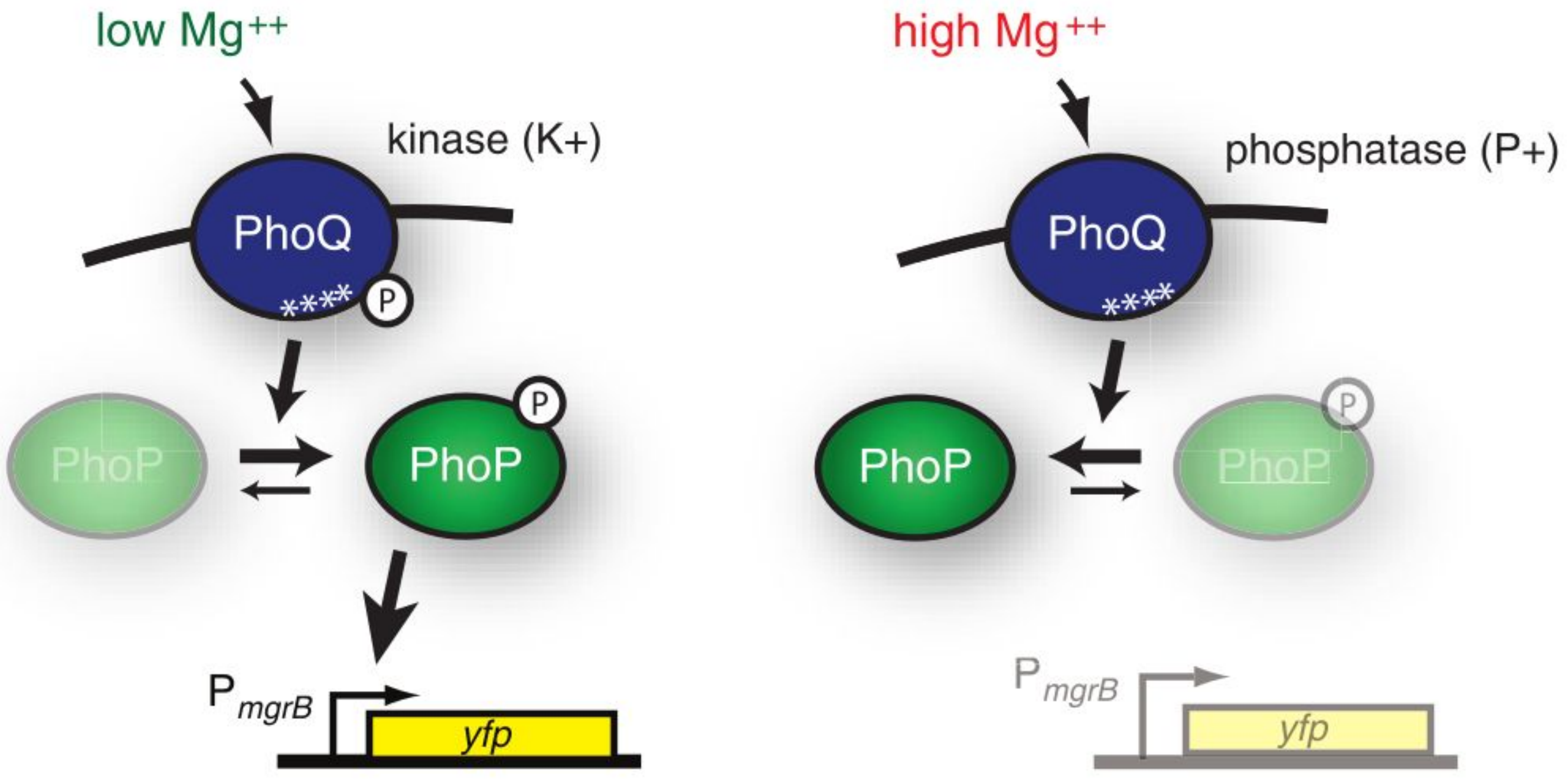
Adapted from Fowler *et al.* (2023) *Genome Biol.*

Understanding basic evolutionary processes



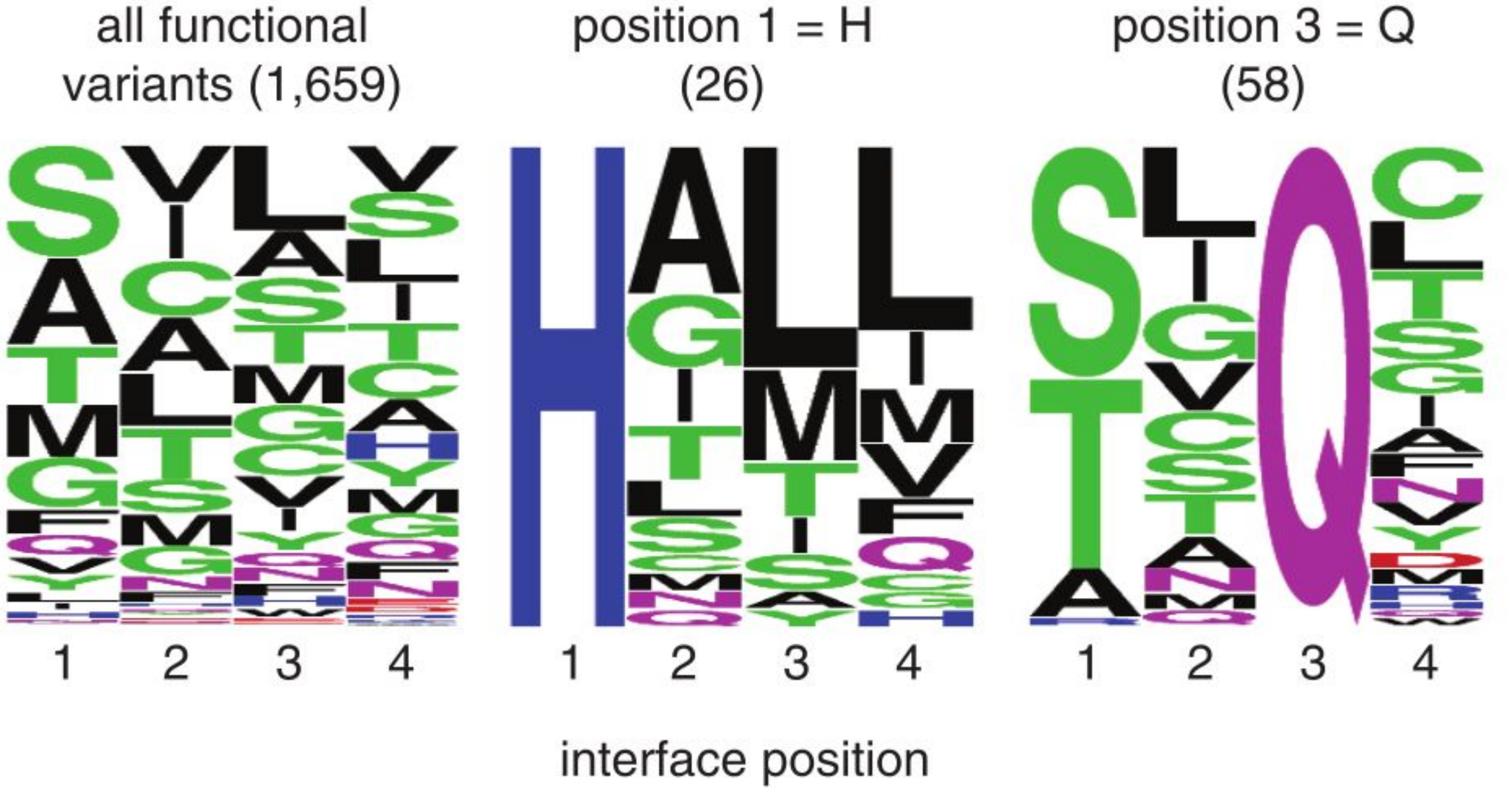
Adapted from Steinberg and Ostermeier (2016) *J Mol Biol.*

Understanding epistasis and evolutionary landscapes



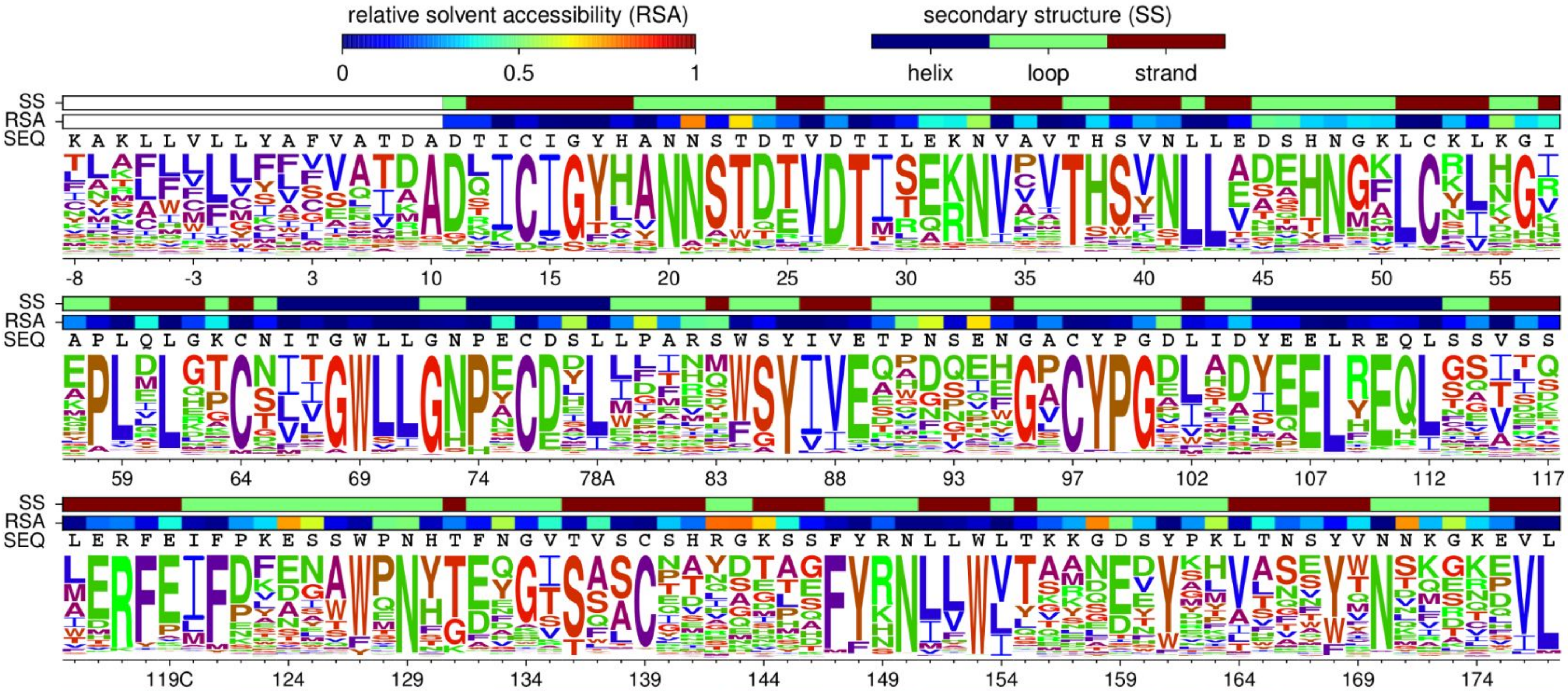
Adapted from Podgornaia and Laub (2015) *Science.*

Understanding epistasis and evolutionary landscapes



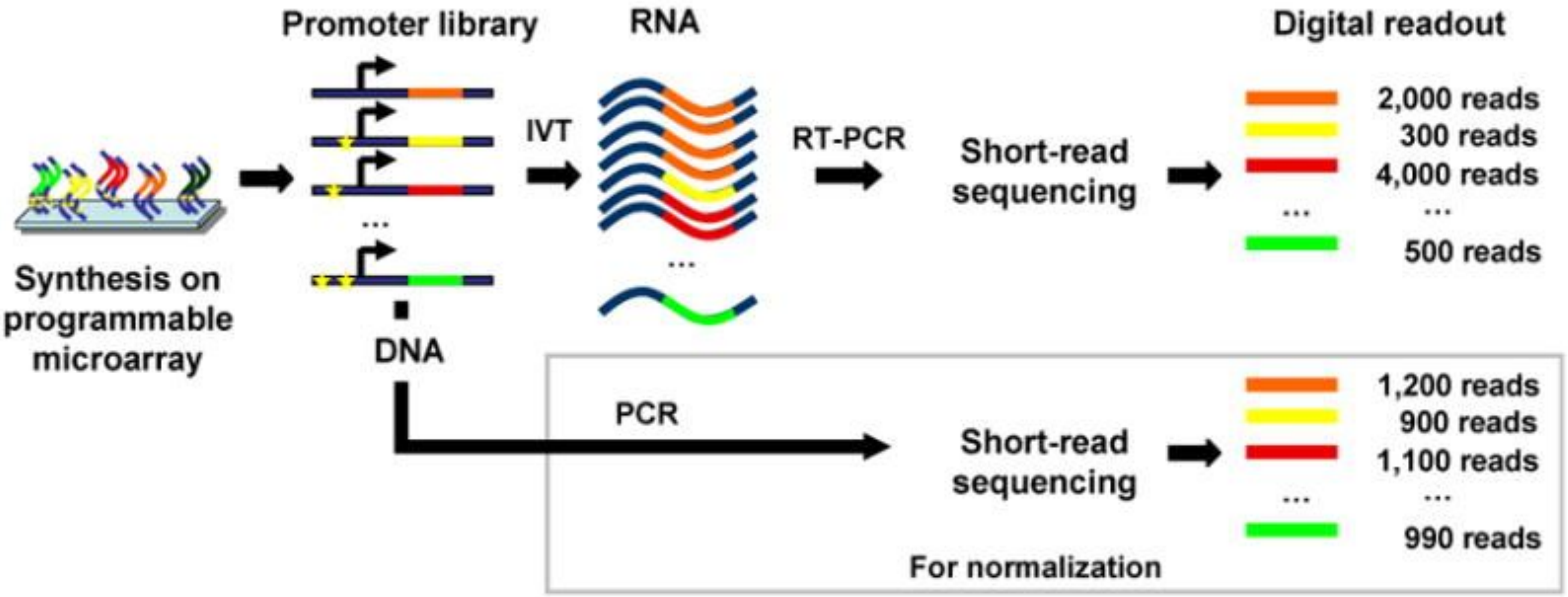
Adapted from Podgornaia and Laub (2015) *Science.*

Interpreting viral sequences



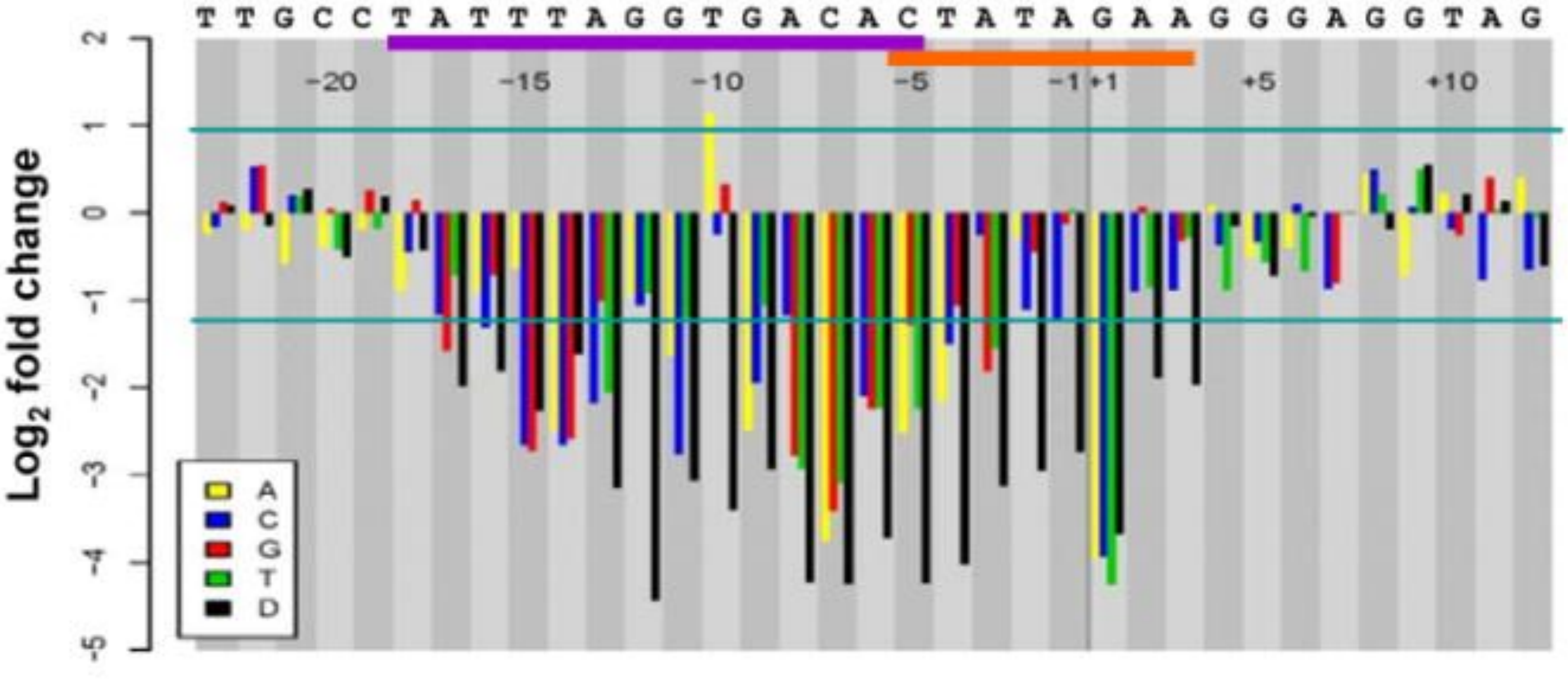
Adapted from Doud and Bloom (2016) *Viruses.*

Non-coding MAVEs



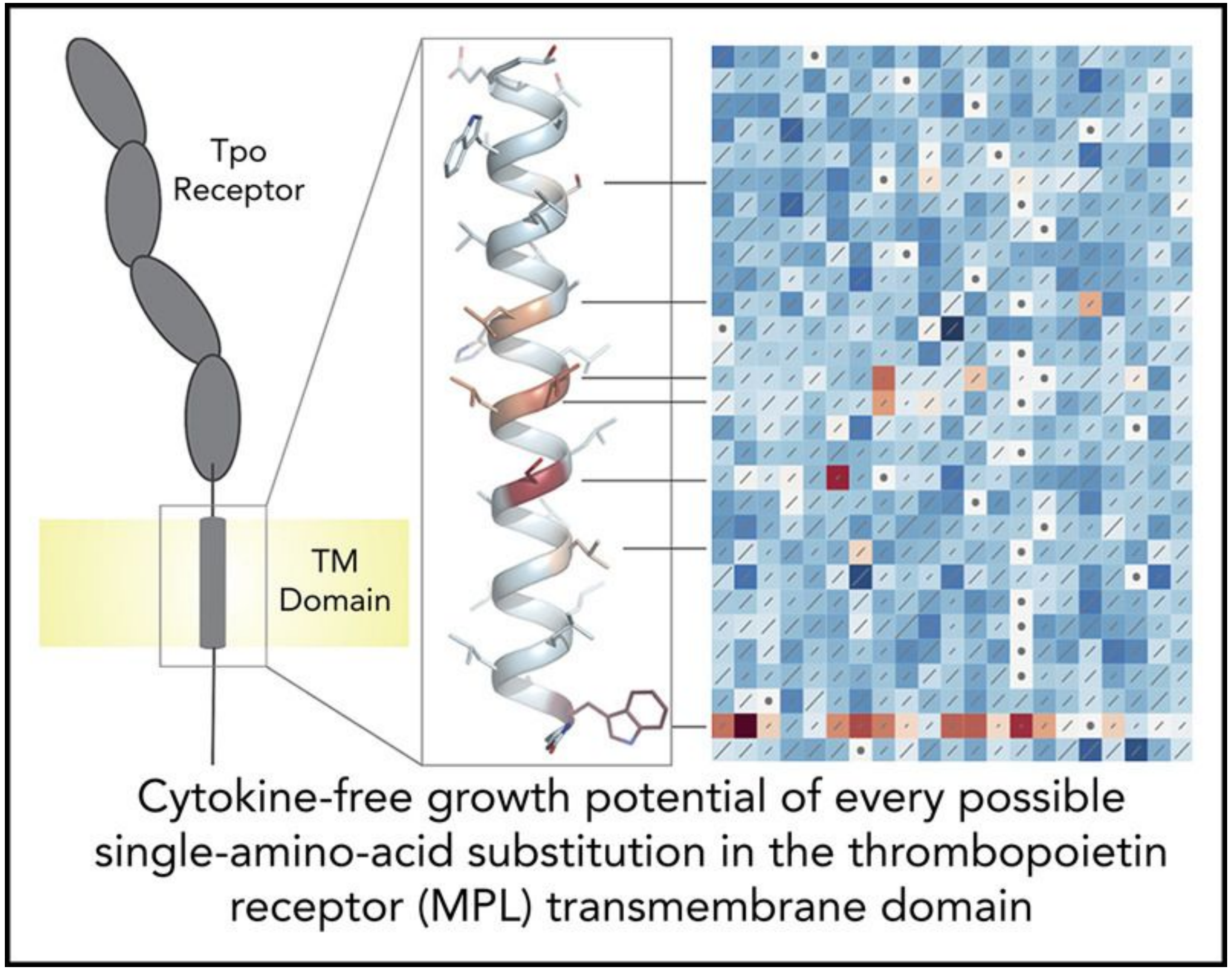
Adapted from Patwardhan *et al.* (2009) *Nat Biotechnol.*

Non-coding MAVEs



Adapted from Patwardhan *et al.* (2009) *Nat Biotechnol.*

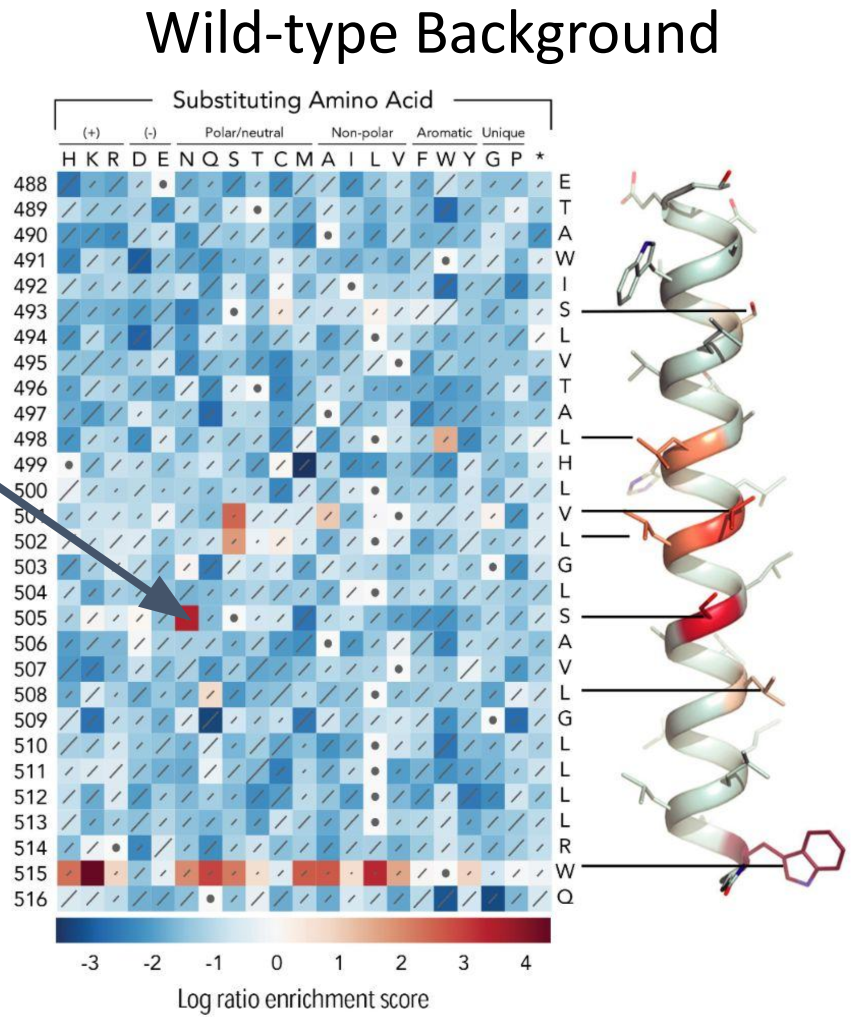
Using MAVEs to understand cancer drivers



Adapted from Bridgford *et al.* (2020) *Blood.*

Using MAVEs to understand cancer drivers

S505N

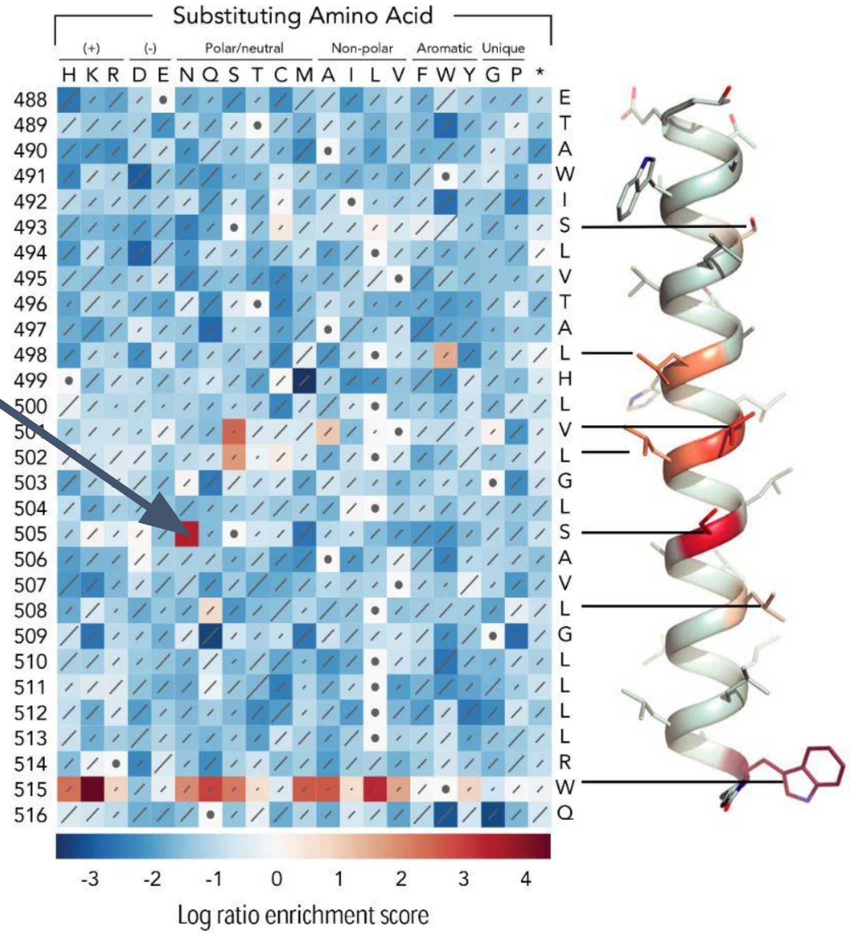


Adapted from Bridgford *et al.* (2020) *Blood.*

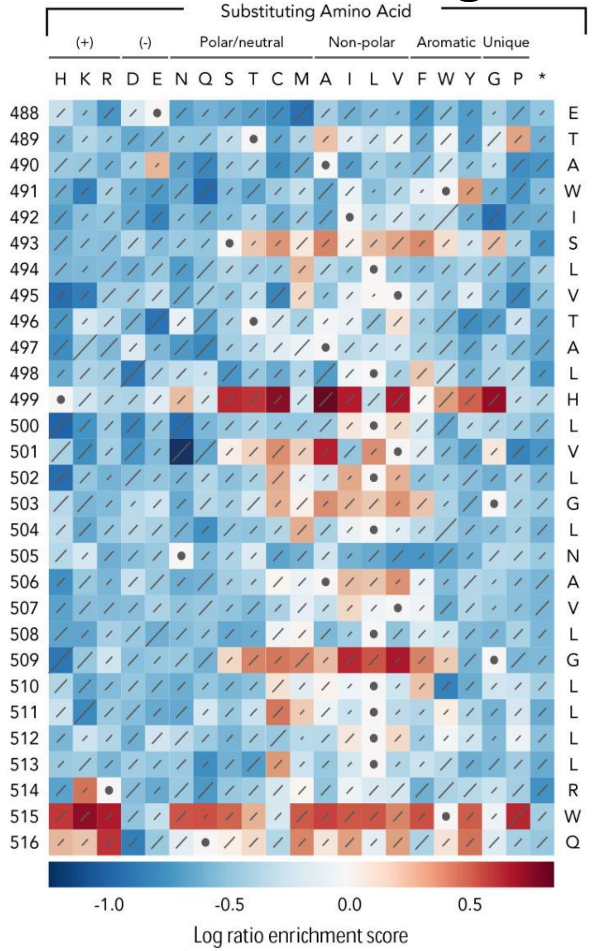
Using MAVEs to understand cancer drivers

Wild-type Background

S505N



S505N Driver Background

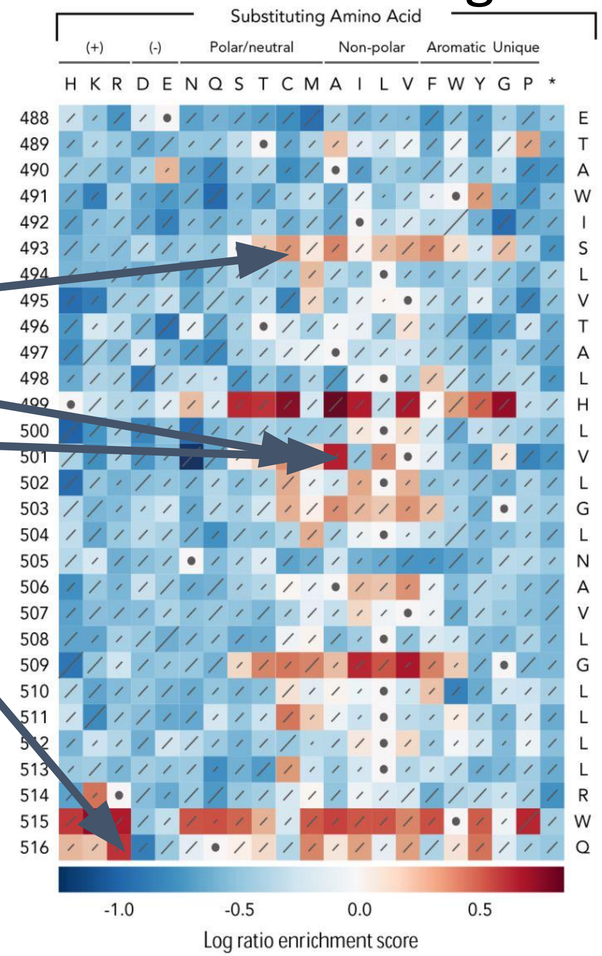


Adapted from Bridgford *et al.* (2020) *Blood.*

Using MAVEs to understand cancer drivers

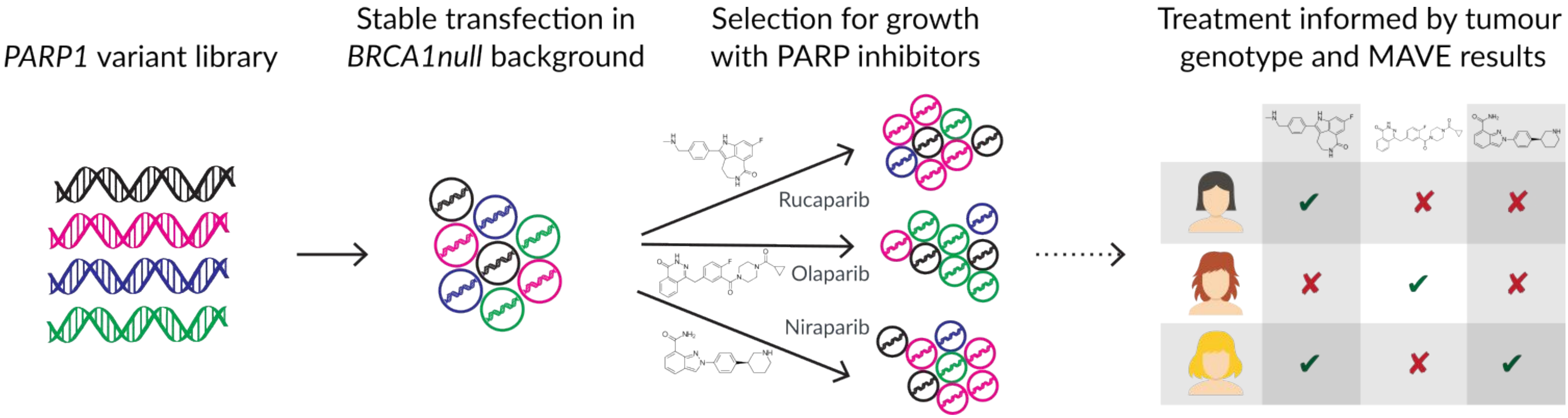
Age (y) at Diagnosis	Gender (M/F)	WHO Diagnosis	MPL Exon 10 Mutation(s)
62	F	Post-ET MF	S505N^{Som}+T487A^{Som}
77	M	ET	S505N^{Som}+S493C^{Som}
61	M	ET	S505N^{Ger}+V501M^{Som}
62	M	Post-ET MF	S505N^{Som}+V501A^{Som}
62	F	Post-ET MF	S505N^{Ger}+Q516R^{Som}
86	M	ET	S505N+V501M+A506V*
57	F	ET	W515L^{Som}+S505C
56	M	ET	W515L^{Som}+V501A
80	F	ET	W515R^{Som}+V501A

S505N Driver Background



Adapted from Bridgford *et al.* (2020) *Blood.*

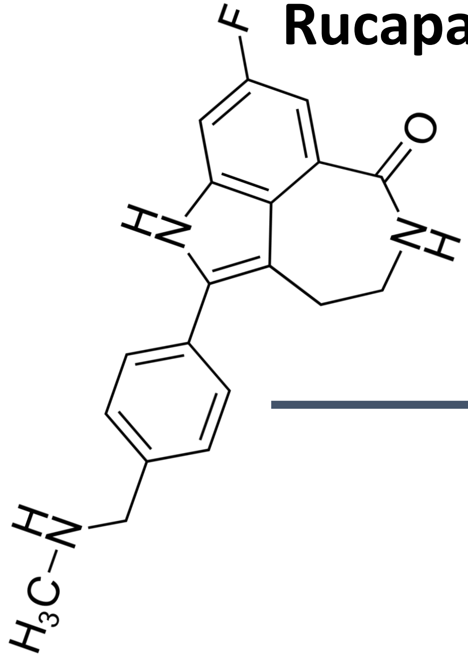
MAVEs for cancer pharmacogenomics

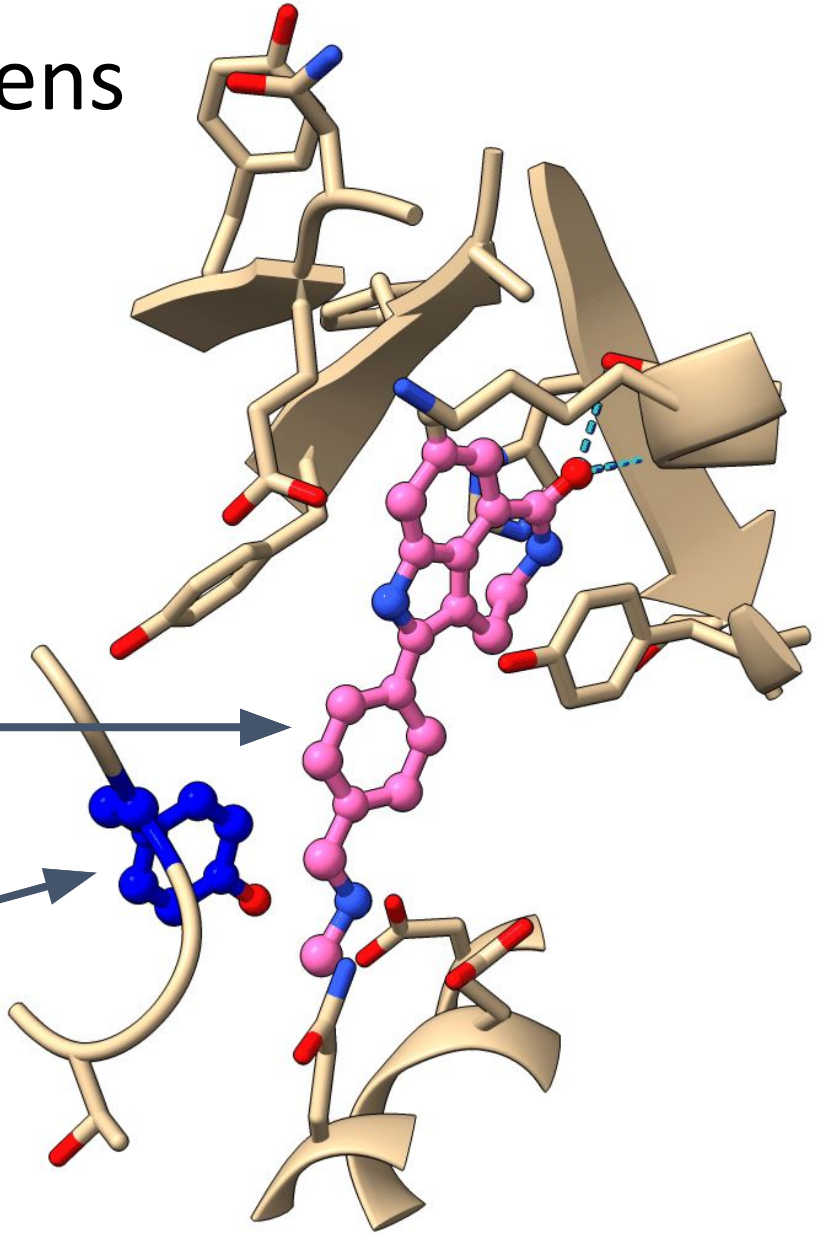


Slide credit Matthew Wakefield

Comparing PARP1+PARPi MAVE screens identifies drug specific resistance

Rucaparib



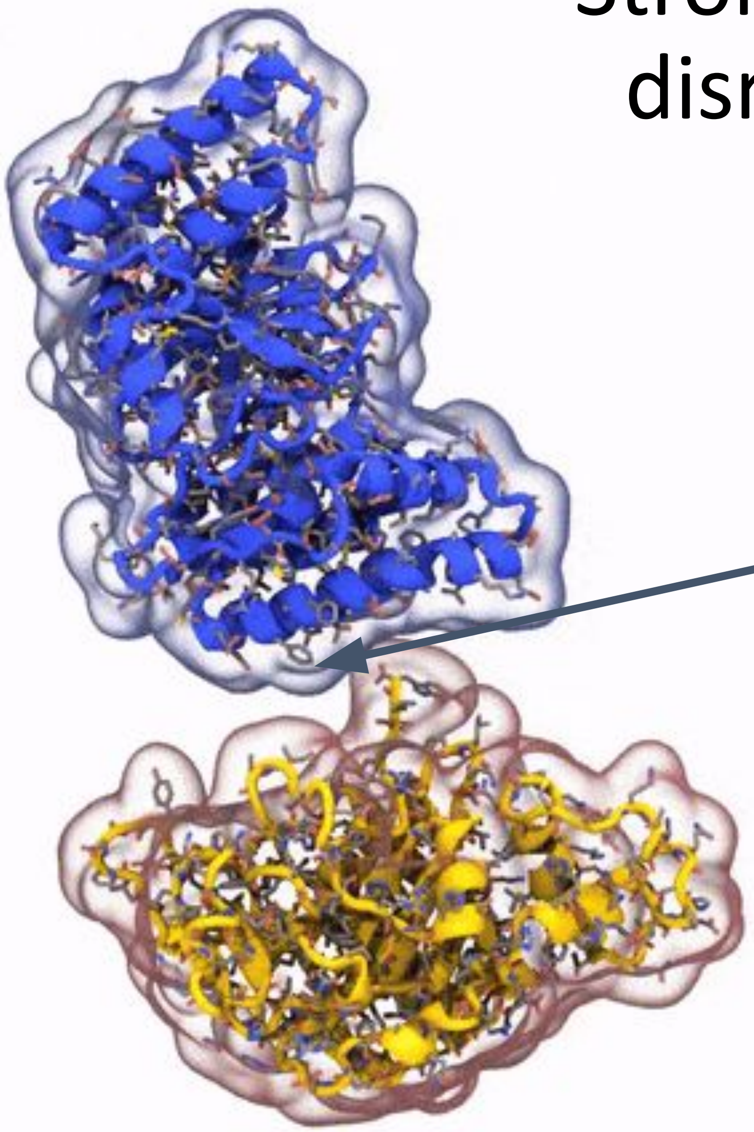


PARP1:p.Tyr889Arg
Strongest Rucaparib-specific variant

Slide credit Matthew Wakefield

PARP1 with Rucaparib Xray structure 4RV6. Only residues <5Å from Rucaparib displayed

Strongest Olaparib-specific variant disrupts HPF1 interaction surface



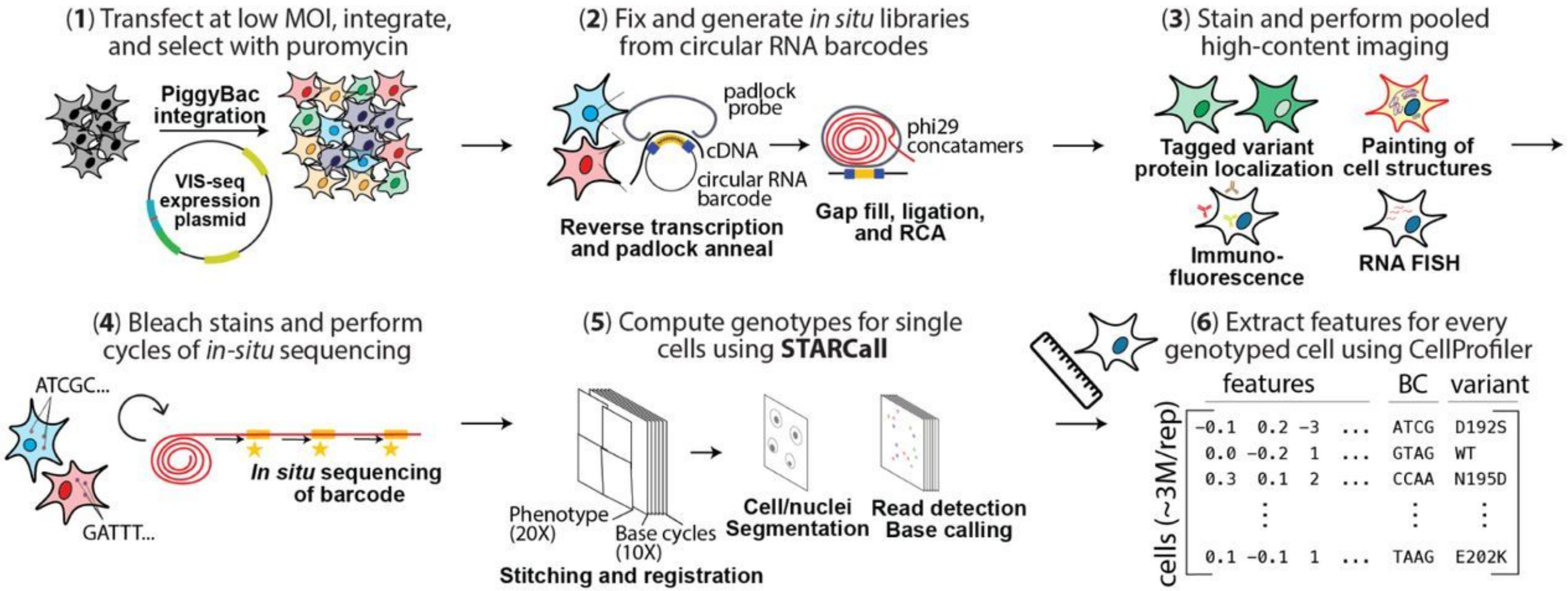
PARP1:p.Thr1010Gln
Increased side chain length 4aa from C-terminus

Olaparib shows HFP1-dependent increased trapping in cell free assays – Rucaparib does not

Slide credit Matthew Wakefield

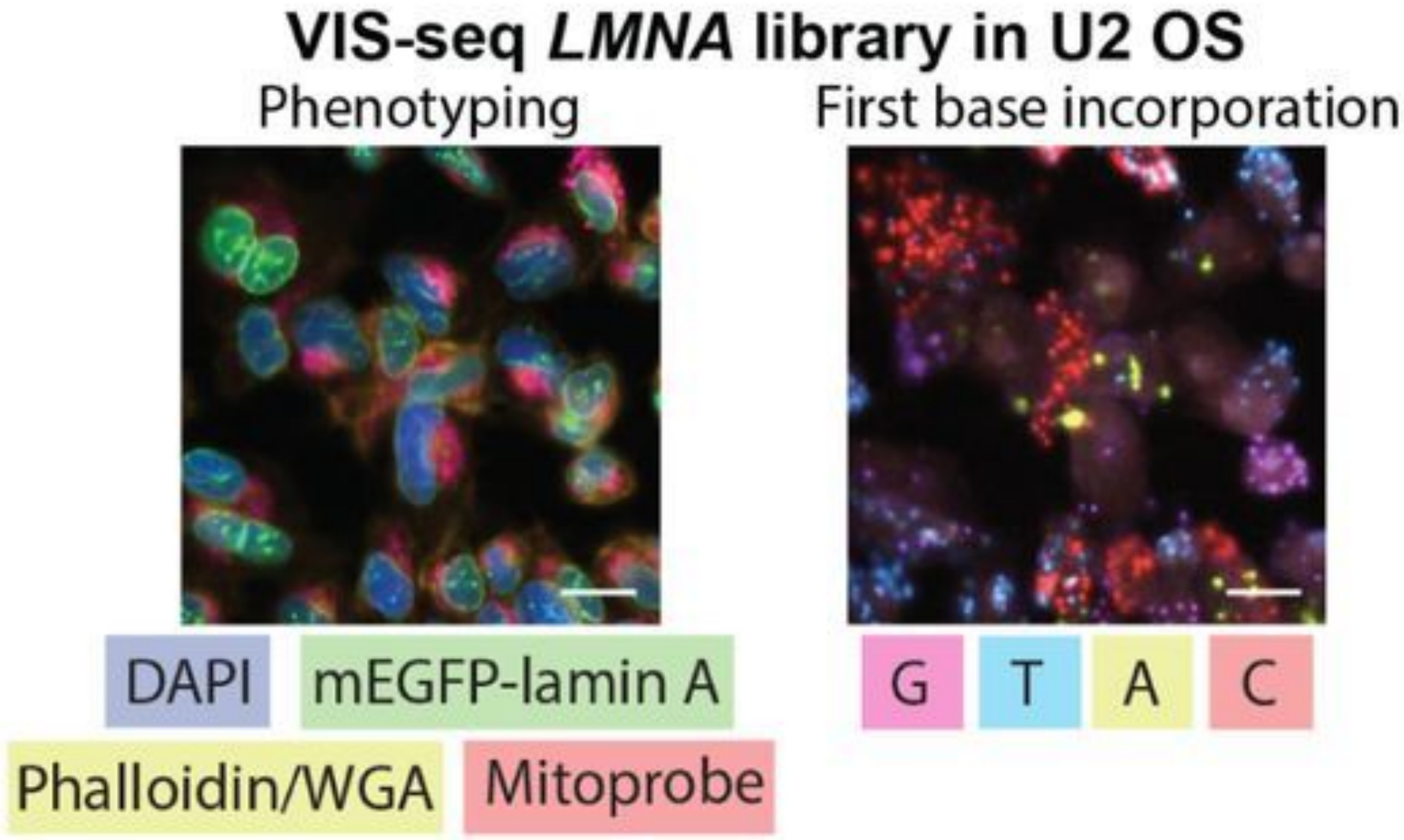
Namd Molecular Dynamics Simulation by Mike Kuiper, Data61 CSIRO

Variant in situ sequencing (VIS-seq)



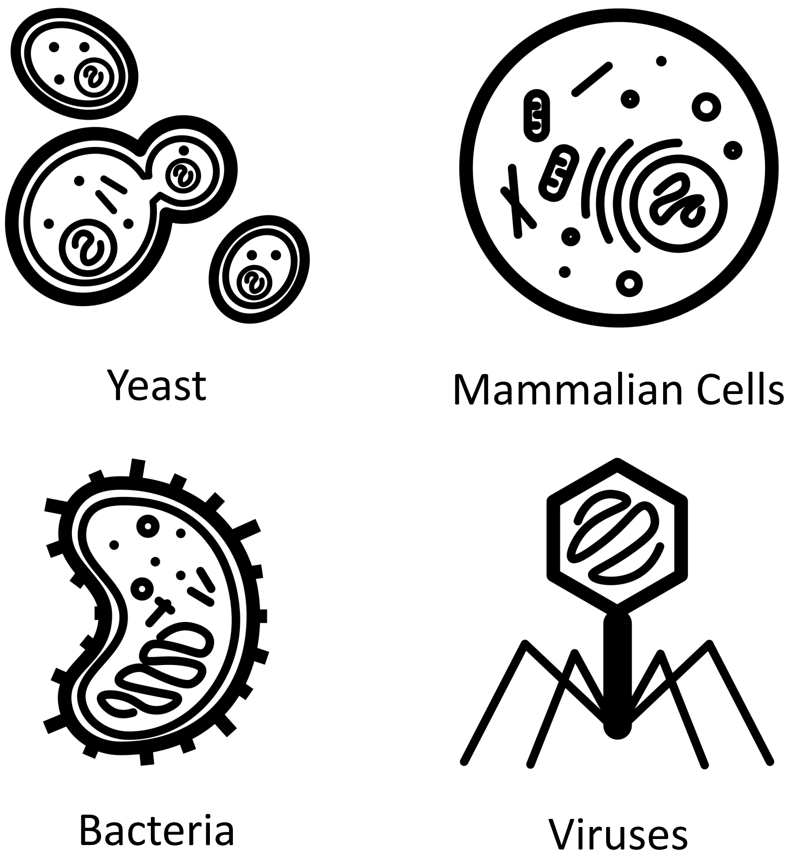
Adapted from Pendyala *et al.* (2025) *bioRxiv.*

Variant in situ sequencing (VIS-seq)

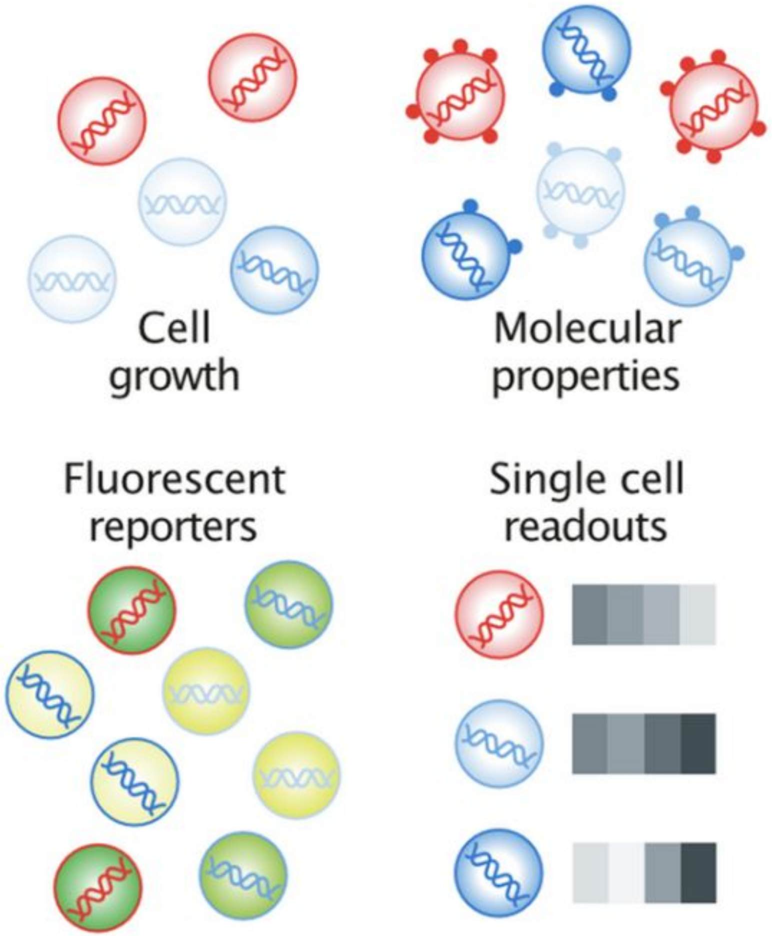


Adapted from Pendyala *et al.* (2025) *bioRxiv.*

Heterogeneity is a major challenge for functional data



Assay System



Assay Readout

Experiments and data formats are heterogeneous

	A	B	C	D	E
1		sequence	name	100 nm Bcl-xL	100 nm Mcl-1
2	A1	MRPEIWIAQELRRIGDEFNAYYARRVZ	Bim	102158715.91	48726484.67
3	A2	MRPEIWIAQELRRIGDEVNAYYARRVZ	LIB2	44719093.27	101847831.9
4	A3	MRPEIWIAQELRRIGDENNAYYARRVZ	LIB3	33112634.6	109079610.26
5	A4	MRPEIWIAQELRRFGDEFNAYYARRVZ	LIB4	134182581.26	30231775.88
6	A5	MRPEIWIAQELRRFGDEVNAYYARRVZ	LIB5	36231804.96	100832032.79
7	A6	MRPEIWIAQELRRFGDENNAYYARRVZ	LIB6	34541032.83	35650206.41
8	A7	MRPEIWIAQELRRDGDEFNAYYARRVZ	LIB7	118089888.27	7233422.72
9	A8	MRPEIWIAQELRRDGDEVNAYYARRVZ	LIB8	19042519	35121970.04
10	A9	MRPEIWIAQELRRDGDENNAYYARRVZ	LIB9	13144871.33	5978852.54
11	A10	MRPEIWIAQELRRNGDEFNAYYARRVZ	LIB10	140189071.17	8712365.88
12	A11	MRPEIWIAQELRRNGDEVNAYYARRVZ	LIB11	44008865.14	58978529.28

	A	B	C	D
1	Nucleotide Position	Nucleotide Mutation	Amino Acid Substitution	RF index
2	30	A30T	Q7L	0.037281726
3	32	T32C	C8R	0.120064898
4	33	G33A	C8Y	0.869537172
5	34	C34T	C8C	0.426665334
6	35	T35C	F9L	0.11946419
7	36	T36C	F9S	0.078118763
8	36	T36A	F9Y	0.326153588
9	38	A38G	N10D	0.157902285

	A	B	C	D	E	F	G	H
1	Pos	Genotype	Env	type	DNA_1	DNA_2	virus_1	virus_2
2	1	A1C	I1L	missense	0	0	0.00033	0
3	1	A1G	I1V	missense	0.00081	0.00107	0.03002	0.02591
4	1	A1T	I1F	missense	0.00115	0.00069	0.00134	0.00109
5	2	T2A	I1N	missense	0.00037	0.00044	0.00031	0.00047
6	2	T2C	I1T	missense	0.00087	0.00075	0.00443	0.00279
7	2	T2G	I1S	missense	6.20E-05	5.01E-05	0	0
8	3	C3A	I1I	silent	0.00015	0.00011	0.00654	0.00465
9	3	C3G	I1M	missense	3.10E-05	5.01E-06	0.00074	0.00047

	A	B	C	D	E	F
1	***Apparent Fitness Landscape of the MS2 Coat Protein***					
2	**Residue #**	**A**	**S**	**T**	**V**	**C**
3	**1**	-0.18	-0.25	0.84	-0.15	Not Observe
4	**2**	0.44	0.67	1.14	-0.19	-4.00
5	**3**	-1.63	-1.18	0.83	0.49	-4.00
6	**4**	-1.33	-0.13	-0.90	-0.42	-0.23
7	**5**	0.56	0.03	-0.31	0.62	0.14
8	**6**	0.86	0.58	0.84	-1.39	-0.48
9	**7**	-1.66	-0.56	-0.82	-0.38	-0.81
10	**8**	0.09	-0.42	0.64	0.05	-4.00

F A I R

Findable Accessible Interoperable Reusable

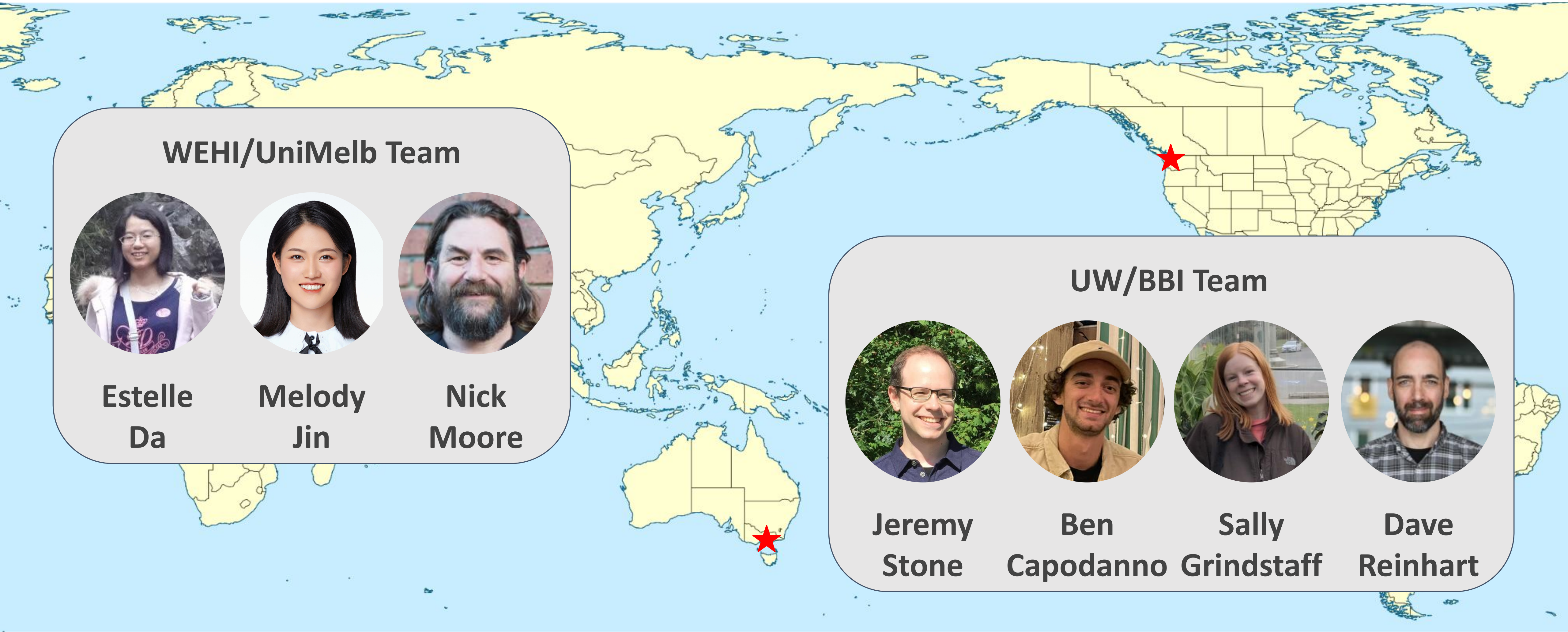
Adapted from Wilkinson *et al.* (2016) *Sci Data*.

MaveDB

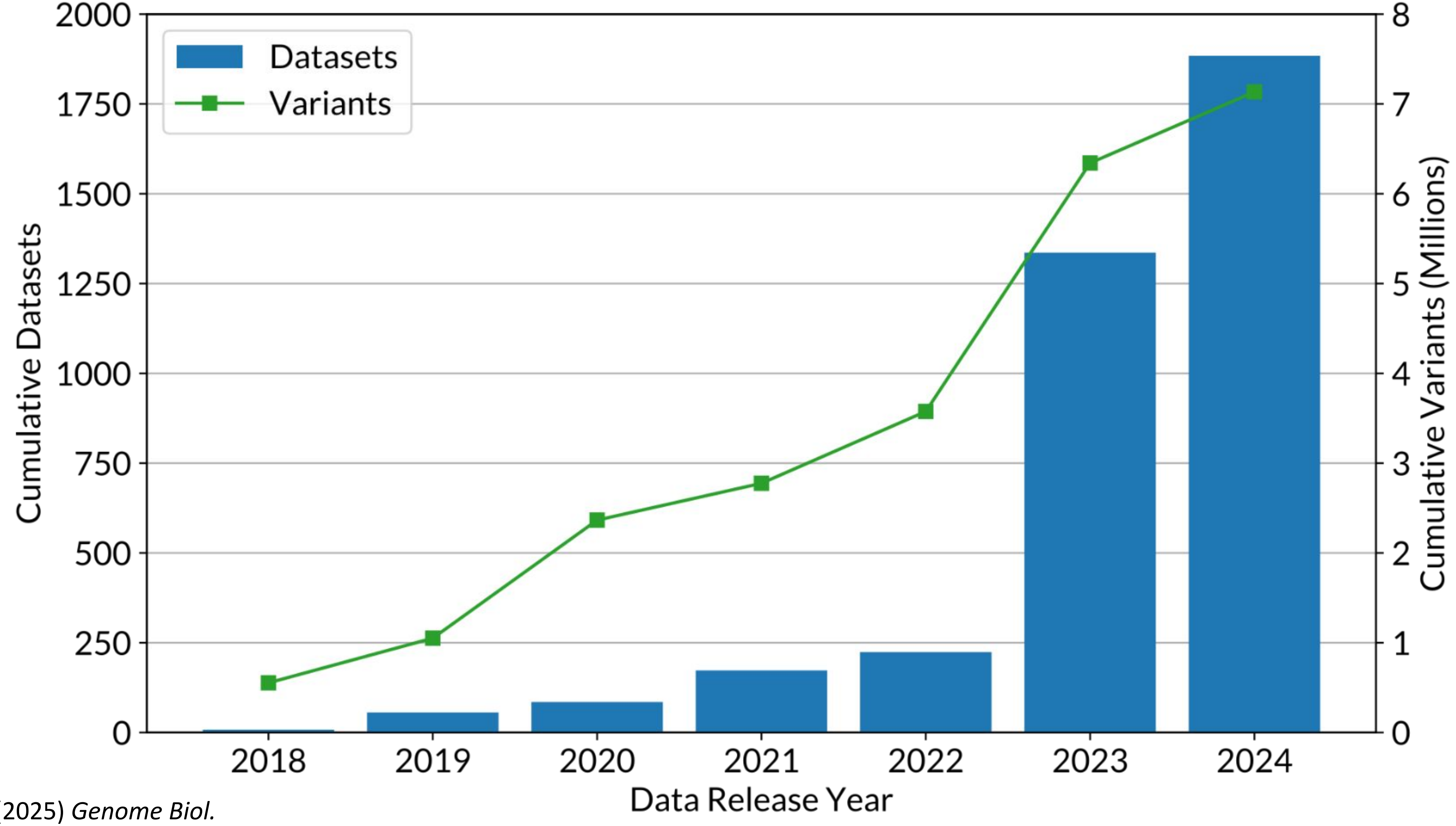
- MaveDB is the database for multiplexed functional data
- Launched in 2019
  - Esposito et al. *Genome Biol.*
- Contains ~800 datasets and 8M variant measurements
- https://www.mavedb.org

Logo designed by Sayeh Gorjifard

MaveDB is built by an international team

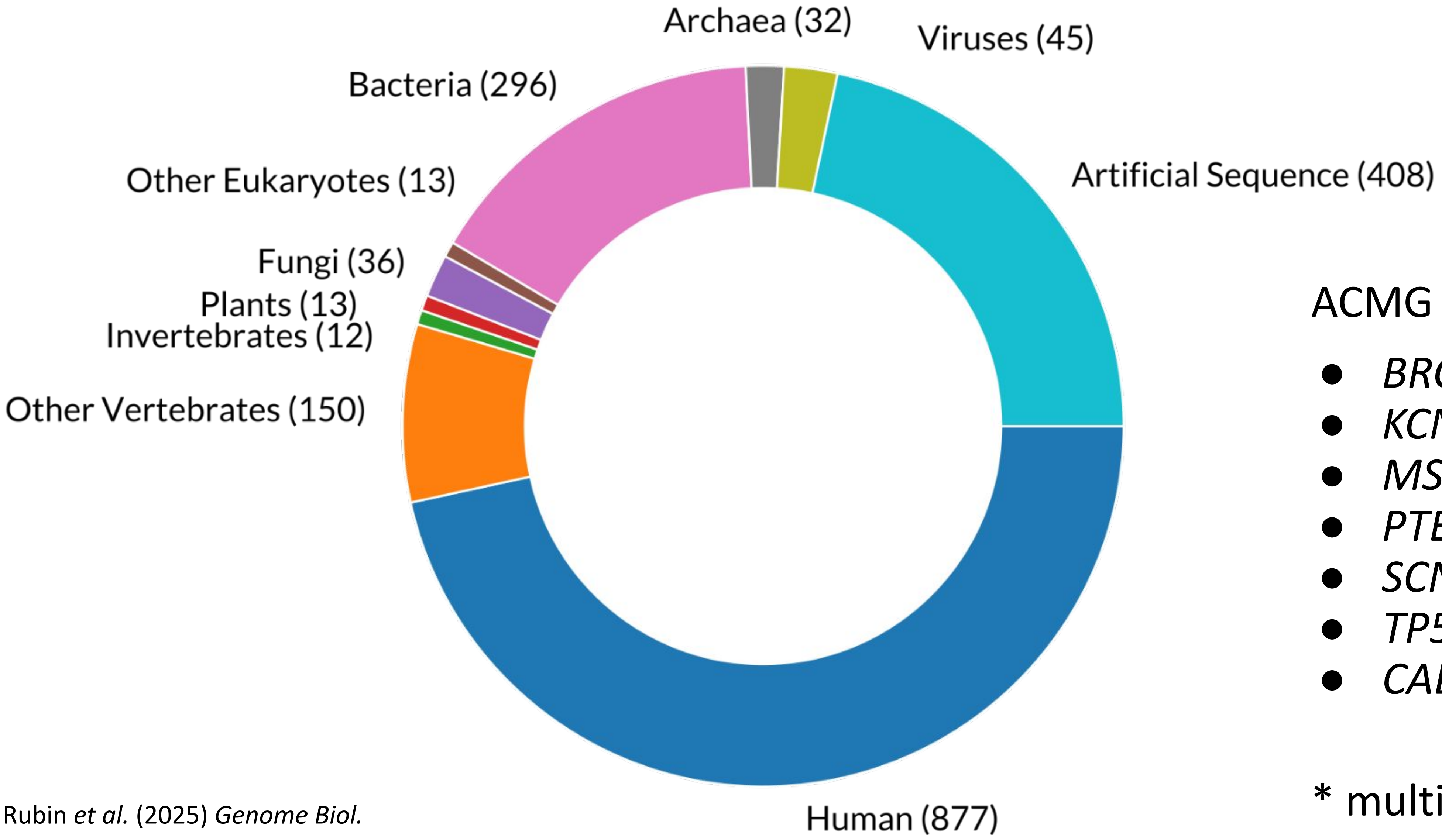


Millions of variants are available in MaveDB



Rubin *et al.* (2025) *Genome Biol.*

MaveDB data are diverse but mostly human



ACMG genes in MaveDB:

- *BRCA1**
- *KCNQ1*
- *MSH2*
- *PTEN**
- *SCN5A*
- *TP53**
- *CALM1/2/3*

* multiple datasets

Rubin *et al.* (2025) *Genome Biol.*

MaveDB web interface

Automatically-generated visualizations

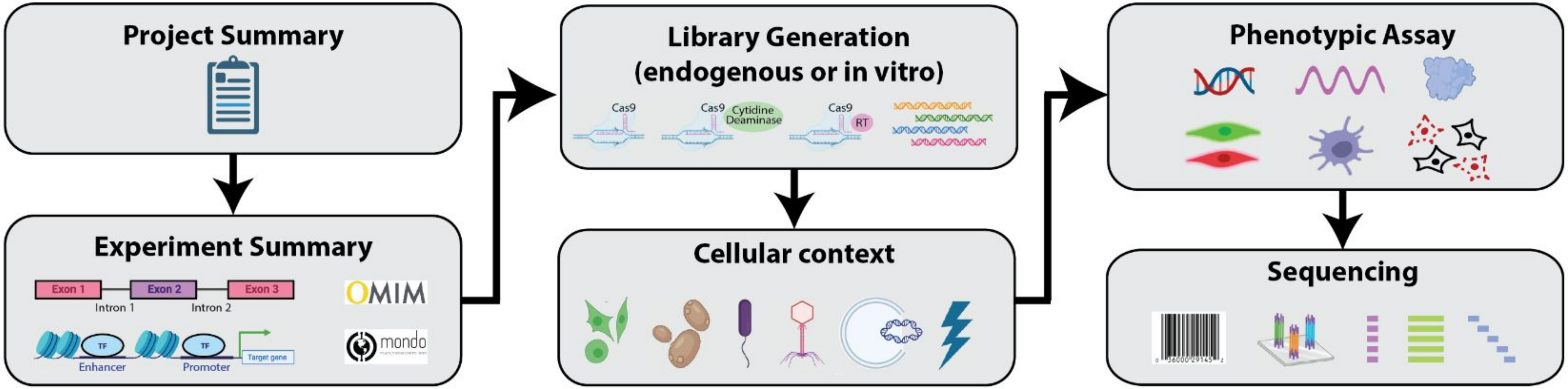
Linkouts to other data resources

Abstract, methods and publication details

Target details

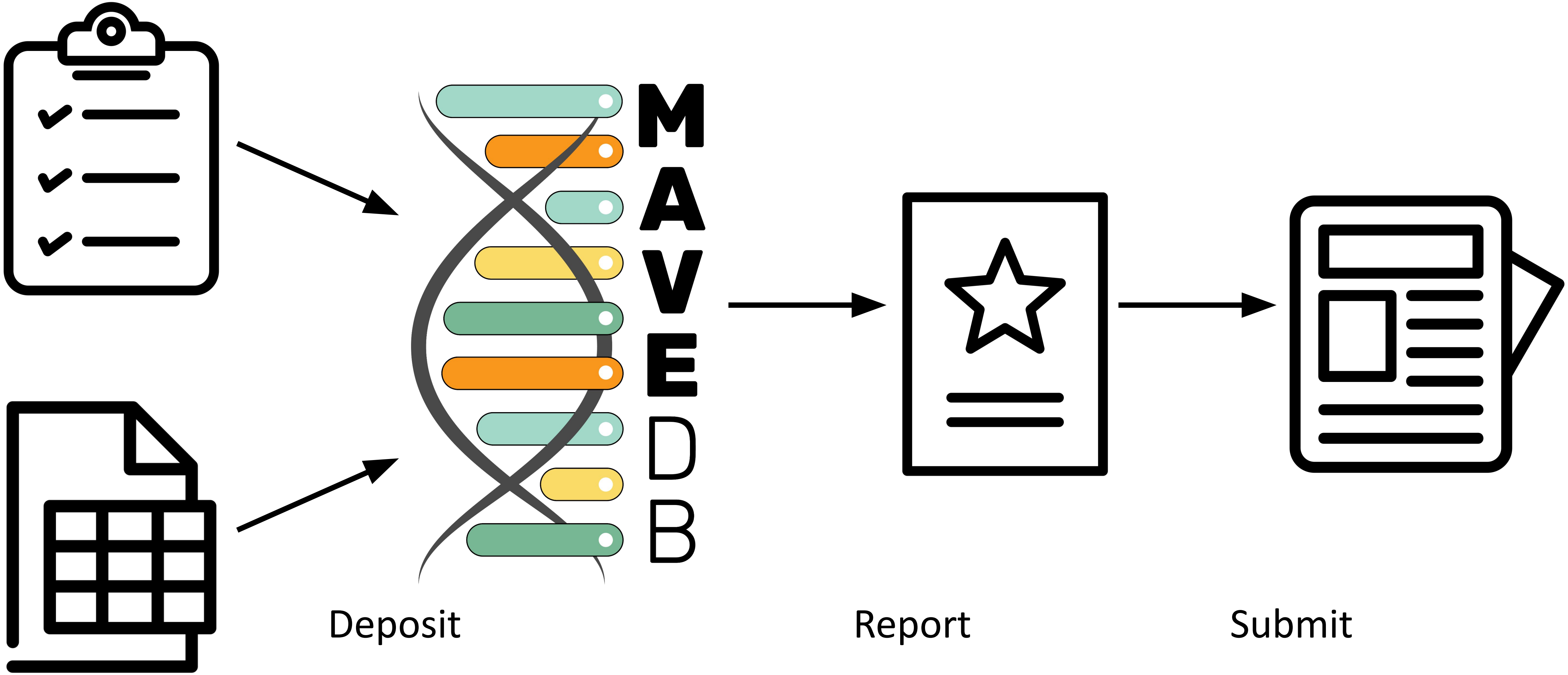
Data table preview

Minimum information standards and a controlled vocabulary

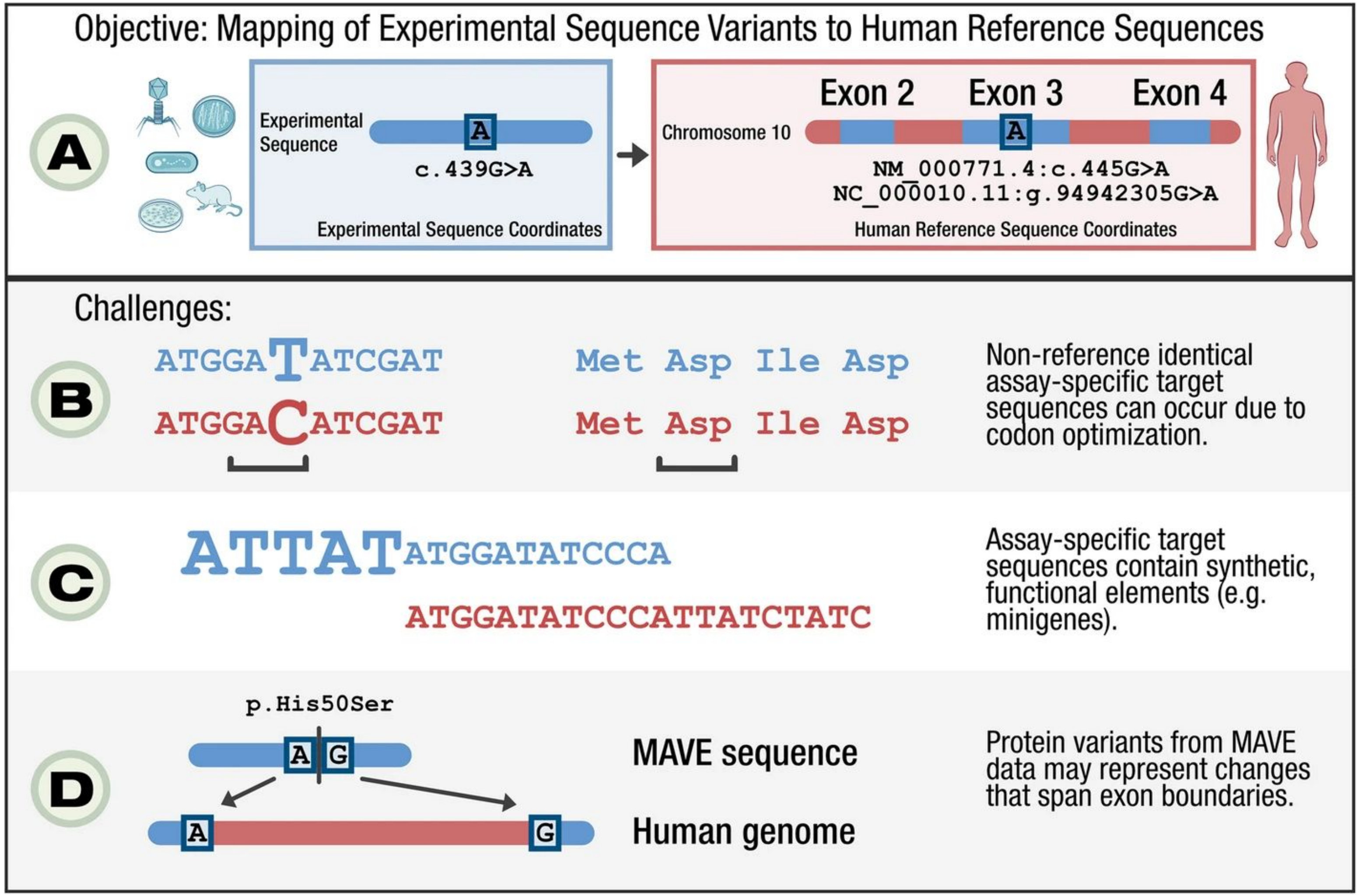


Adapted from Claussnitzer, Parikh, Wagner, *et al.* (2025) *Genome Biol.*

Using standards to promote data deposition



Mapping MaveDB variants to the human genome



Arbesfeld *et al.* (2025) *Genome Biol.*

MaveDB implements GA4GH standards

MAVEDB

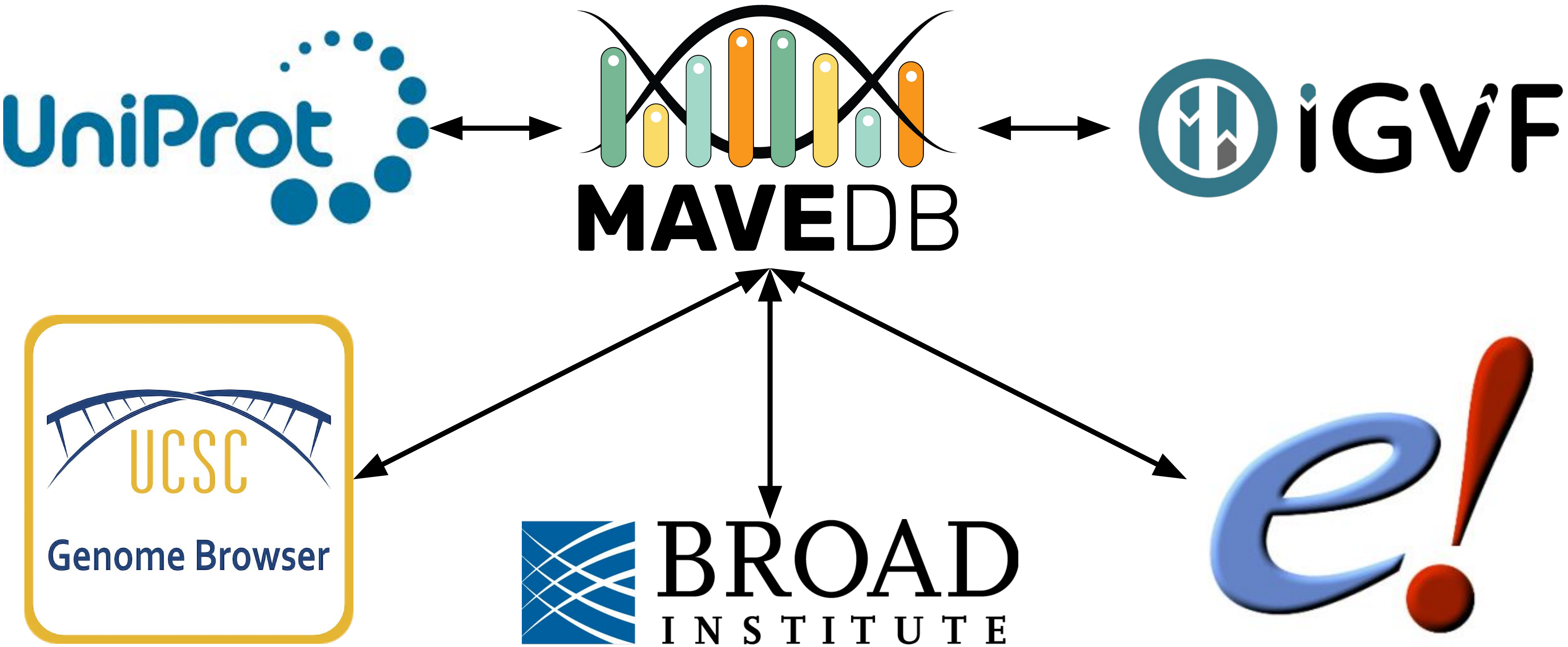
GA4GH

ClinVar
Clinically relevant variation

ClinGen
Clinical Genome Resource

DECIPHER
GRCh38

Encouraging contributions through data integration



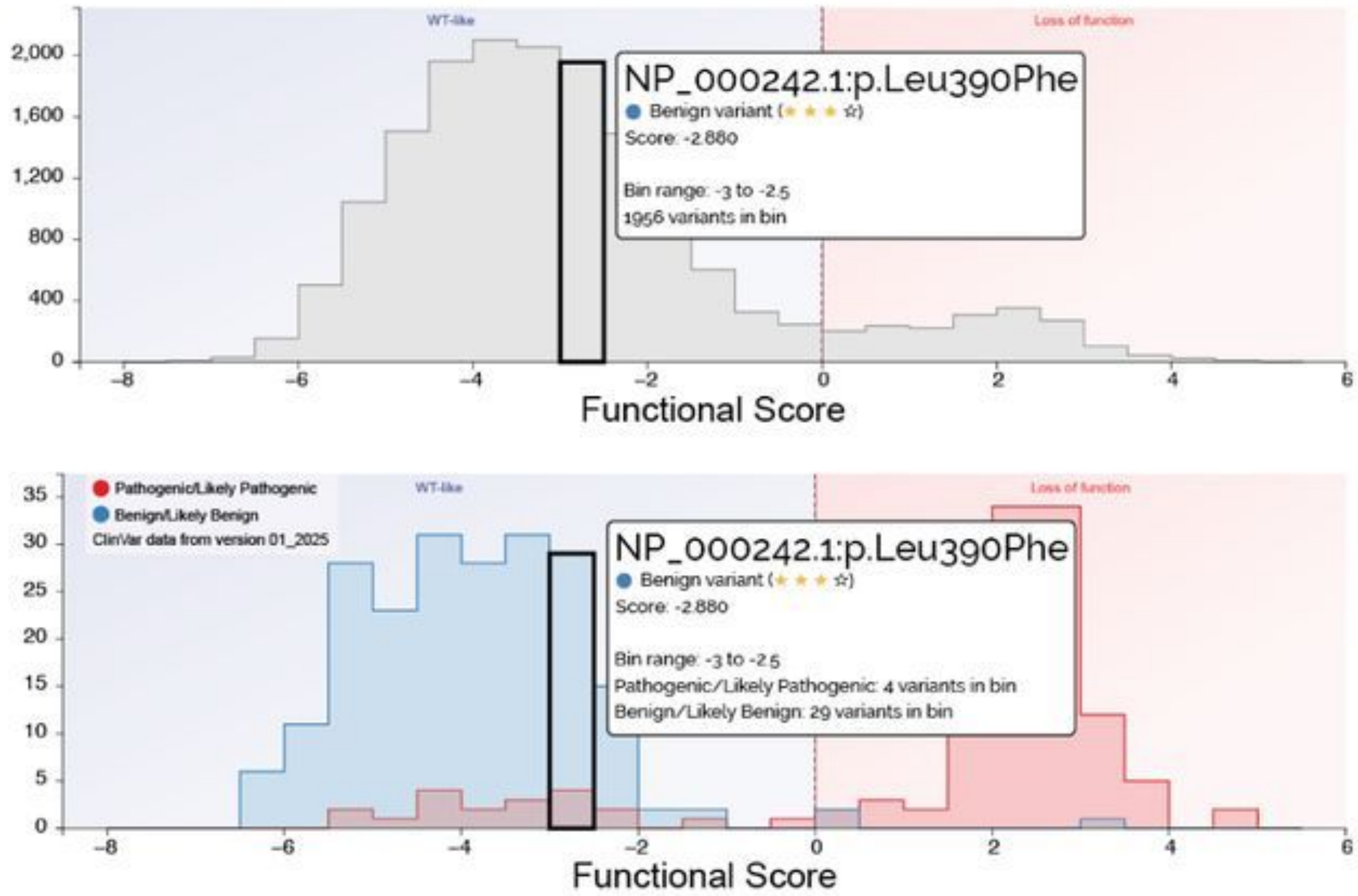
New MaveMD clinical interface

Jia et al. MSH2 2021

Gene (HGNC symbol)	MSH2	
Assay Type	Cell fitness	
Molecular Mechanism Assessed	DNA repair, mismatch repair	
Variant Consequences Detected	Loss of function	
Model System	Immortalized human cells (HAP1)	
Detects Splicing Variants?	No	
Detects NMD Variants?	No	
Number of Variants	17,746	
Clinical Performance		
OddsPath – Normal	0.0483	BS3_Strong
OddsPath – Abnormal	198	PS3_Strong

Adapted from McEwen, *et al.* (2025) *medRxiv.*

New MaveMD clinical interface



Adapted from McEwen, *et al.* (2025) *medRxiv.*

New MaveMD clinical interface

Investigator-provided functional classes		
Baseline Score: -4.00		
Range label	WT-like	Loss of function
Classification	Normal	Abnormal
Numeric Interval	(-∞, 0.00]	(0.00, ∞)
Evidence strength	BS3_STRONG	PS3_STRONG
OddsPaths ratio	0.0483	198
Thresholds: Jia X et al. (2021) — **Method:** Brnich SE et al. (2019)		

Adapted from McEwen, *et al.* (2025) *medRxiv.*

Acknowledgements

WEHI Bioinformatics
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- Eva Chan

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- Estelle Da
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Atlas of Variant Effects
- Jeremy Arbesfeld
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- Dustin Baldridge
- Kevin TA Booth
- Alicia Byrne
- Jeffrey Calhoun
- Rui Chen
- Randy Cockerell
- Shawn Fayer
- Andrew Glazer
- Anna L Gloyn
- Jenny Goldstein
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- Lea Starita
- Patricia Sullivan
- Clare Turnbull
- Rehan Villani
- Barbara Vona
- Alex Wagner
- Daniel Zimmerman

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Wellcome Sanger Institute

Prof Maitreya Dunham
University of Washington

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Kyoto University

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UC Santa Cruz

Amelie Stein
University of Copenhagen

Alex Wagner
Nationwide Children's Hospital

Jian-Rong Yang
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National Speakers

Melissa Call
WEHI

Mark Dawson
Peter MacCallum Cancer Centre

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Chai-Ann Ng
Victor Chang Cardiac Research Institute

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The Royal Melbourne Hospital

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